Ref. [36] — Introductory Lectures on Convex Optimization

Y. Nesterov | Kluwer / Springer, 2004 (monograph)

1. Why this book appears in our non-convex survey

The book develops the mathematical baseline for first-order optimization and oracle complexity, including nonlinear, smooth convex, nonsmooth convex and structural optimization. Our survey cites it as *background*, not as a paper that proves saddle escape in deep neural networks. We need its definitions and the descent calculations that remain valid for smooth *non-convex* functions, while keeping convex-only conclusions separate.

2. From first principles: smoothness and a descent step

Suppose f has L-Lipschitz gradient:

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|.$$

The smoothness/descent lemma gives for any increment s

$$f(x+s) \le f(x) + \langle \nabla f(x), s\rangle + \frac{L}{2}\|s\|^2.$$

Taking the gradient step $s = -\eta\nabla f(x)$,

$$f(x - \eta\nabla f(x)) \le f(x) - \eta\left(1 - \frac{L\eta}{2}\right)\|\nabla f(x)\|^2.$$

For $\eta = 1/L$, this simplifies to

$$f(x_{t+1}) \le f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2.$$

This calculation *does not require convexity*: it only needs global L-smoothness.

3. Derive the basic non-convex stationarity rate

If f is bounded below by $f_{\inf}$, sum the previous inequality over T steps:

$$\frac{1}{2L}\sum_{t=0}^{T-1}\|\nabla f(x_t)\|^2 \le f(x_0) - f(x_T) \le \Delta, \quad \Delta = f(x_0) - f_{\inf}.$$

Consequently,

$$\boxed{\min_{0\le t<T}\|\nabla f(x_t)\|^2 \le \frac{2L\Delta}{T}.}$$

To obtain a gradient norm no larger than ϵ, it suffices to take $T \ge 2L\Delta/\epsilon^2$. This is a *teaching derivation* of the standard first-order bound; it should not be mistaken for a deep-network global-optimality theorem by the book.

4. Convexity changes the meaning of the gradient

Convexity provides the extra inequality

$$f(y) \ge f(x) + \langle \nabla f(x), y - x\rangle.$$

Therefore, if $\nabla f(x^\star) = 0$, then $f(y) \ge f(x^\star)$ for every y; stationarity is global optimality. Non-convex functions lack that implication. For $f(x) = x^3$ one has $f'(0) = 0$ but 0 is neither a local nor a global minimum. This simple counterexample explains the transition from convex textbook guarantees to the approximate-stationarity targets of Sections IV–V of our survey.

5. Acceleration and complexity: correctly delimit the context

Nesterov-type acceleration is foundational for improving rates of smooth *convex* minimization. It does *not* follow that the same convex proof establishes efficient escape from non-convex saddles. That later claim belongs to specialized works (e.g., our Ref. [10]) whose accelerated updates include additional curvature-sensitive

analysis or interventions. The distinction matters when discussing how optimization rates depend on what the algorithm can query.

What this reference contributes to the survey. A baseline framework of smoothness, first-order methods, mathematical rates and oracle-based comparisons. It supports the notation and the elementary $O(\epsilon^{-2})$ first-order stationarity calculation, without asserting second-order stationarity or deep-learning generalization.

6. Scope and source limitation

The publisher confirms the 2004 monograph and its four substantive chapters. The complete book text was not available for line-by-line checking here. Accordingly, these derivations are *explicitly pedagogical reconstructions of standard optimization facts*, not numbered quotations, verified theorem numbers, or a claim that the full book was read. They can serve as background notes, but a theorem-by-theorem book review would require the text.

Survey relevance (Preliminaries/RQ4). Cite this book for classical first-order optimization background, not for a theorem about arbitrary non-convex neural-network minima or efficient saddle escape.

7. Our notes

This is the mathematical reference behind the basic gradient descent and smoothness arguments. The key lesson for our survey is which parts survive without convexity: loss decrease and small-gradient guarantees can survive, while "small gradient means global minimum" does not. **Publisher:** https://doi.org/10.1007/978-1-4419-8853-9.